

MACHINE LEARNING

REGRESSION FRAMEWORK

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Regression: the stochastic framework

The **stochastic/population** model

- ▶ We have a quantitative response Y and p predictors $X_1, \dots, X_p$.
- ▶ We assume there is a relationship between Y and $X = (X_1, \dots, X_p)$

$$Y = f(X) + \epsilon,$$

where ϵ is a random **error term** with $E(\epsilon) = 0$ and $\text{Var}(\epsilon) = \sigma^2$.

- ▶ The function f is fixed but **unknown** and represents the **systematic** information that X provides about Y .

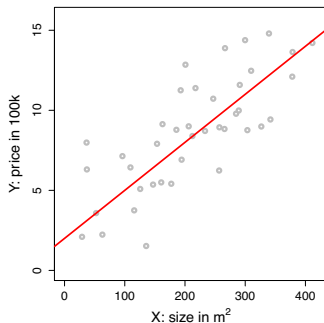
Example: [$Y = \text{house price (100k)}$; $X = \text{size (m}^2\text{)}$]

$$Y = 2 + 0.03X + \epsilon$$

$$\text{house price} = 2 + 0.03 \times \text{size} + \text{error}$$

- ▶ This true relationship $f(X) = 2 + 0.03X$ is called the **population regression line**.
- ▶ Note: (X, Y) is a $(p + 1)$ -dimensional random vector. In this simulated example:

$$X = \mathcal{N}(0, \sigma_X^2), \text{ and } Y = 2 + 0.03X + \mathcal{N}(0, \sigma_\epsilon^2).$$



Regression: the statistical framework

The training data

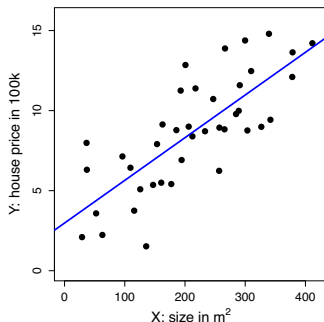
- ▶ Assume we are given n measurements $(x_1, y_1), \dots, (x_n, y_n)$ of (X, Y) , where $x_i = (x_{i1}, \dots, x_{ip})^\top$, and
 - ▶ the inputs $x_i \in \mathbb{R}^p$ are called **predictors**;
 - ▶ the outputs y_i are called **responses**.
- ▶ We will use this data to **learn** the unknown relationship f between Y and X , i.e., to **train** (or **estimate**) a predictive model $\hat{f}$.

Example: [$Y = \text{house price (100k)}$; $X = \text{size (m}^2\text{)}$]

$$\text{house price} \approx \hat{f}(\text{size}) = 2.977 + 0.027 \times \text{size}$$

- ▶ The training data are realizations of the random vector (X, Y) , e.g.,

	X	Y
(x_1, y_1)	327	9.0
(x_2, y_2)	154	7.9
(x_3, y_3)	233	8.7
(x_4, y_4)	206	9.0
(x_5, y_5)	96	7.1
(x_6, y_6)	257	6.2
...	...	...



Why do we estimate f ?

1. Prediction

- ▶ A set of inputs X are readily available, but Y cannot be easily obtained.
- ▶ We can predict Y by

$$\hat{Y} = \hat{f}(X),$$

where $\hat{f}$ is our **predictive model** for f .

- ▶ The function $\hat{f}$ is often treated as **black box**: we are not concerned with the exact form of $\hat{f}$ provided it yields good prediction for Y .

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2. Inference

- ▶ We are often interested in understanding **how** Y is affected as $X_1, \dots, X_p$ change.
- ▶ Cannot treat $\hat{f}$ as black box, we need to know the exact form.
- ▶ We want to answer questions such as:
 - ▶ Which predictor is associated with the response?
 - ▶ What is the relationship of the response and each predictor (positive, negative,...)?
 - ▶ Is the relationship linear or more complicated?
 - ▶ etc.

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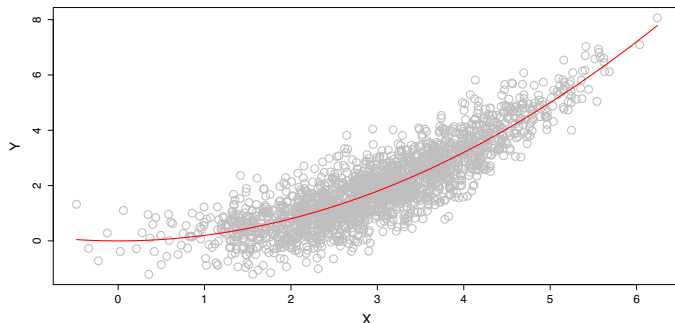
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2. Inference (Classical multivariate statistics)

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 - ▶ etc.

What is a good $\hat{f}$?



- ▶ For a fixed value of X , say $X = 3$ is there an **optimal prediction** $\hat{f}(X)$?
- ▶ There might be many values at $X = 3$, and a good choice, minimizing squared error, is

$$\hat{f}(3) = E(Y \mid X = 3),$$

where $E(Y \mid X = 3)$ is the **expected value** (average) of Y given $X = 3$.

The loss function

- ▶ We have to specify what we mean with a **good prediction**. This will depend on the application.
- ▶ Introduce a **loss function** L that measures the discrepancy between a response y and its prediction $\hat{y}$.
- ▶ In the regression setup, we typically use **squared error loss**, i.e.,

$$L(y, \hat{y}) = (y - \hat{y})^2.$$

- ▶ For our predictive model $\hat{f}$, the **expected squared prediction error** is

$$\text{Err}_{\hat{f}} = \mathbb{E}[\{Y - \hat{f}(X)\}^2],$$

where the expectation is taken over (X, Y) .

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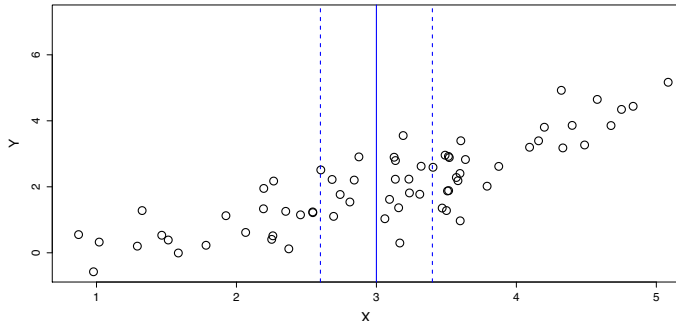
- ▶ Ideally, we would like to find the optimal f^* that **minimizes** Err_{f^*} .
- ▶ In fact, f^* is the so-called **regression function**

$$\hat{f}(x) \approx f^*(x) = \mathbb{E}(Y|X = x),$$

(X, Y)

which satisfies $f^* = f$ if $Y = f(X) + \epsilon$.

How do we estimate $\hat{f}$?



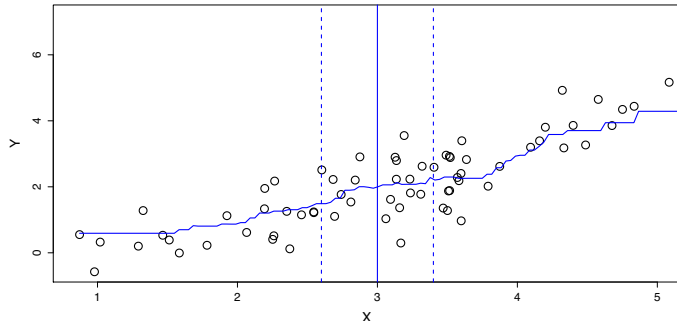
- ▶ Typically we have **few or no** data points with $X = 3$.
- ▶ So we cannot compute $E(Y | X = 3)$!
- ▶ The **k-Nearest-Neighbors** method aims at approximating this expectation by the average

$$\hat{f}(x_0) = \text{Ave}(Y | X \in N_k(x_0)),$$

where $N_k(x_0)$ are the k closest points x_i to $x_0 = 3$.

- ▶ The number k is a **tuning parameter**.

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- ▶ Nearest neighbor averaging works good for small dimension p and a large training data size n .
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- ▶ We will study **structured methods** that
 - ▶ need **less data** to fit;
 - ▶ may be nicely **interpretable**;
 - ▶ but at the price of **lower flexibility**.
- ▶ The difficulty is to find a good **compromise** between flexibility and parsimony.

The empirical loss: model evaluation

How can we **estimate** the $\text{Err}_{\hat{f}} = \mathbb{E}[\{Y - \hat{f}(X)\}^2]$?

The **test data**

- ▶ For model evaluation we require a new, independent test data set $\mathcal{T}_m = \{(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_m, \tilde{y}_m)\}$.
- ▶ For each input $\tilde{x}_i$ we **predict the response** with our model: $\hat{y}_i = \hat{f}(\tilde{x}_i)$.
- ▶ We compare this prediction with the observed output $\tilde{y}_i$.
- ▶ The **empirical prediction error** of our model $\hat{f}$ is the empirical version of $\text{Err}_{\hat{f}}$, namely

$$\text{err}_{\hat{f}} = \frac{1}{m} \sum_{i=1}^m \{\tilde{y}_i - \hat{f}(\tilde{x}_i)\}^2,$$

also called **mean squared error**.

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- ▶ In practice: separate all data into training and test and do **cross-validation** (later).